Module 2: Frontend Single Page Applications

React Core, Forms & Routing

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Agenda

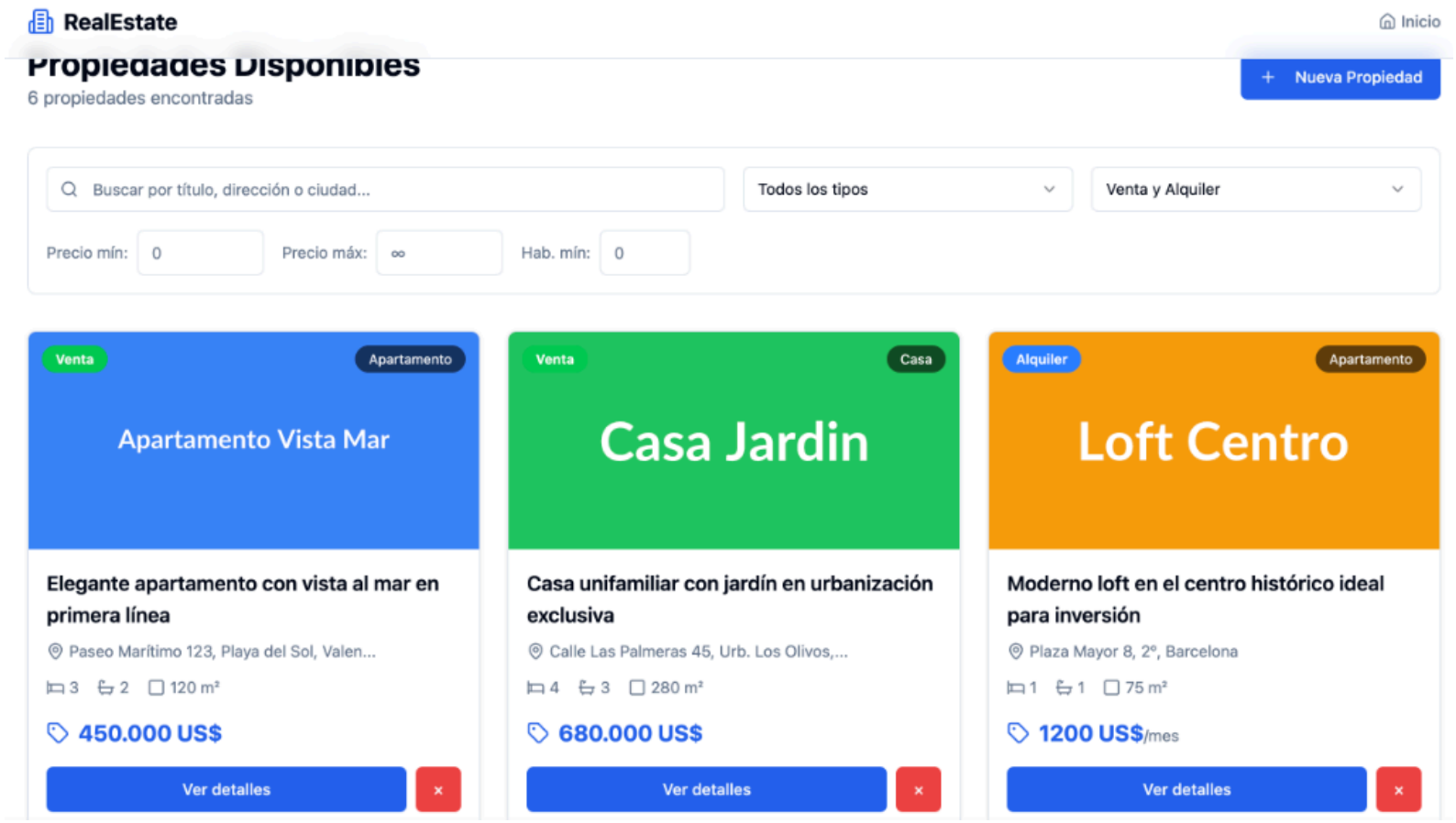
1. **Project: Real Estate React**
2. **React Core (Vite) & Hooks**
3. **Forms (Zod + React Hook Form)**
4. **Routing & LocalStorage**
5. **Deep Dive**
6. **Challenge Lab**

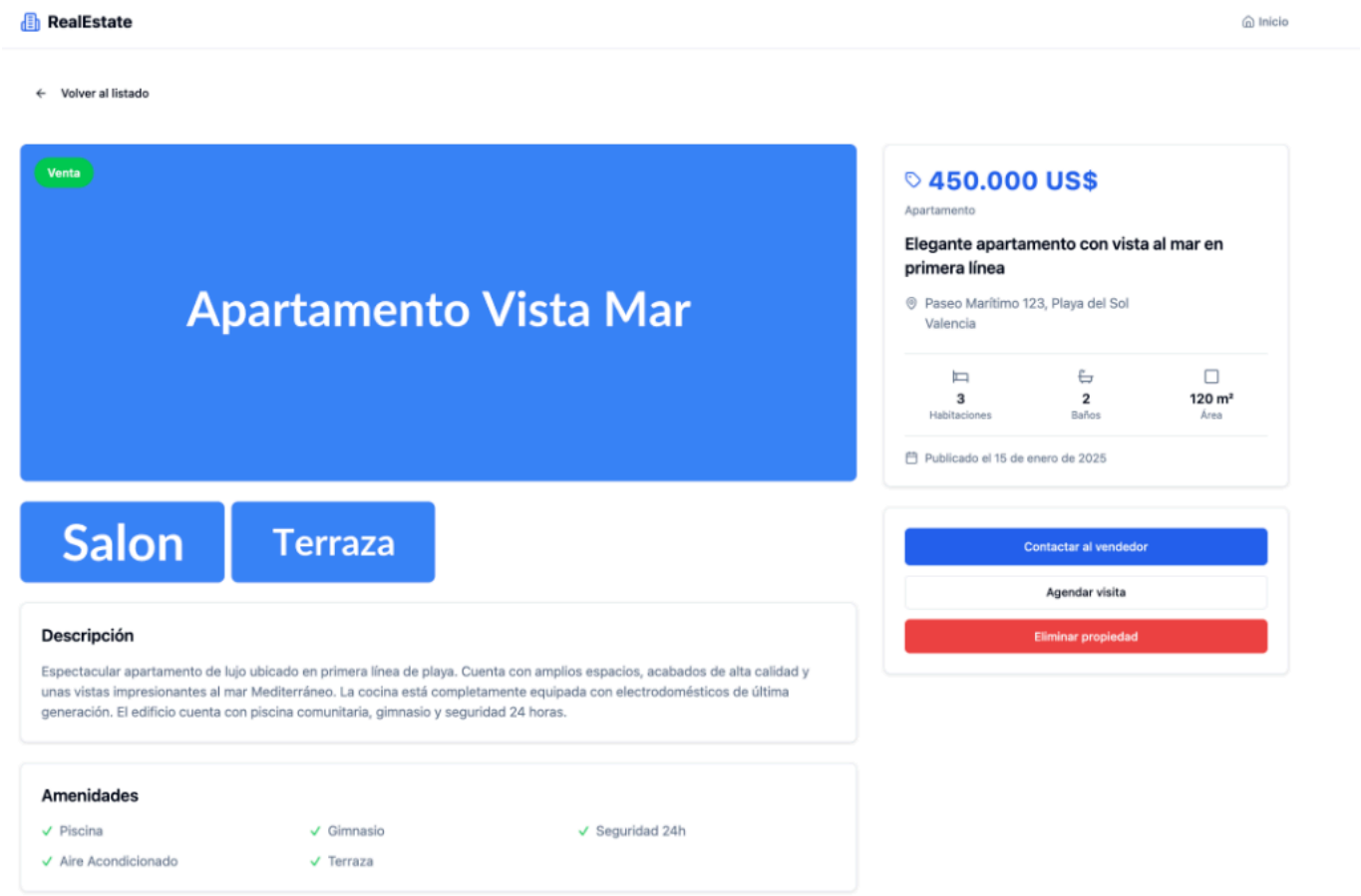
Real Estate App

We are building a Real Estate listing application with React.

Core Requirements:

1. **Listing**: Display property cards with images and details.
2. **Filtering**: Filter by type, price range, and search.
3. **Forms**: Add new properties with validation.
4. **Navigation**: Multi-page app with React Router.


RealEstate
Inicio
Propiedades Disponibles
6 propiedades encontradas
+ Nueva Propiedad
Buscar por título, dirección o ciudad...
Todos los tipos
Venta y Alquiler
Precio mín: 0
Precio máx: ∞
Hab. mín: 0
Venta
Apartamento
Apartamento Vista Mar
Elegante apartamento con vista al mar en primera línea
Paseo Marítimo 123, Playa del Sol, Valen...
3 2 120 m²
450.000 US$
Ver detalles
Venta
Casa
Casa Jardin
Casa unifamiliar con jardín en urbanización exclusiva
Calle Las Palmeras 45, Urb. Los Olivos,...
4 3 280 m²
680.000 US$
Ver detalles
Alquiler
Apartamento
Loft Centro
Moderno loft en el centro histórico ideal para inversión
Plaza Mayor 8, 2º, Barcelona
1 1 75 m²
1200 US$/mes
Ver detalles


RealEstate
Inicio
Volver al listado
Venta
Apartamento Vista Mar
Salon
Terraza
Descripción
Espectacular apartamento de lujo ubicado en primera línea de playa. Cuenta con amplios espacios, acabados de alta calidad y unas vistas impresionantes al mar Mediterráneo. La cocina está completamente equipada con electrodomésticos de última generación. El edificio cuenta con piscina comunitaria, gimnasio y seguridad 24 horas.
Amenidades
Piscina
Gimnasio
Seguridad 24h
Aire Acondicionado
Terraza
450.000 US$
Apartamento
Elegante apartamento con vista al mar en primera línea
Paseo Marítimo 123, Playa del Sol
Valencia
3
Habitaciones
2
Baños
120 m²
Área
Publicado el 15 de enero de 2025
Contactar al vendedor
Agendar visita
Eliminar propiedad

2. React Basics

What is React?

React is a JavaScript library for building user interfaces.

Key Concepts:

1. **Declarative**: Describe what UI should look like, not how to build it.
2. **Component-Based**: Build encapsulated components that manage their own state.
3. **Virtual DOM**: React updates only what changed, not the entire page.

```
// Declarative - What, not how
function Greeting({ name }) {
    return <h1>Hello, {name}!</h1>;
}
```

JSX: JavaScript + HTML

JSX is a syntax extension that looks like HTML in JavaScript.

```
// JSX (What you write)
const element = <h1 className="title">Hello, World!</h1>;

// JavaScript (What it compiles to)
const element = React.createElement(
    'h1',
    { className: 'title' },
    'Hello, World!'
);
```

Key Differences from HTML:

- `className` instead of `class`
- `htmlFor` instead of `for`
- CamelCase for attributes (`onClick`, `onChange`)

Components: The Building Blocks

Components are reusable pieces of UI.

```
// Function Component (Modern)
function PropertyCard({ property }: { property: Property }) {
    return (
        <div className="card">
            <img src={property.imageUrl} alt={property.title} />
            <h2>{property.title}</h2>
            <p>${property.price.toLocaleString()}</p>
        </div>
    );
}

// Usage
<PropertyCard property={myProperty} />
```

Props: Passing Data Down

Props are read-only inputs to components.

```
// Parent component
function App() {
    return (
        <PropertyCard
            title="Beach House"
            price={500000}
            bedrooms={3}
        />
    );
}

// Child component
interface PropertyCardProps {
    title: string;
    price: number;
    bedrooms: number;
}

function PropertyCard({ title, price, bedrooms }: PropertyCardProps) {
    return <div>{title} - ${price} - {bedrooms} beds</div>;
```

Children: Composition

The `children` prop allows component composition.

```
// Reusable Card wrapper
function Card({ children }: { children: React.ReactNode }) {
    return (
        <div className="bg-white rounded-lg shadow-md p-4">
            {children}
        </div>
    );
}

// Usage
<Card>
    <h2>Property Title</h2>
    <p>Description here...</p>
    <Button>View Details</Button>
</Card>
```

Conditional Rendering

Render different UI based on conditions.

```
function PropertyCard({ property }: Props) {
    return (
        <div className="card">
            {/* Conditional with && */}
            {property.featured && <Badge>Featured</Badge>}

            {/* Ternary operator */}
            {property.available
                ? <span className="text-green-500">Available</span>
                : <span className="text-red-500">Sold</span>
            }

            {/* Early return pattern */}
            {!property.imageUrl && <Placeholder />}
        </div>
    );
}
```

Lists & Keys

Render arrays of data with `.map()`.

```
function PropertyList({ properties }: { properties: Property[] }) {
    return (
        <div className="grid grid-cols-3 gap-4">
            {properties.map((property) => (
                <PropertyCard
                    key={property.id}  // Required unique identifier
                    property={property}
                />
            ))}
        </div>
    );
}
```

Why Keys? React uses keys to identify which items changed, were added, or removed.

3. Hooks & State

useState: Component State

State is data that can change over time.

```
import { useState } from 'react';

function Counter() {
    // Declare state variable with initial value
    const [count, setCount] = useState(0);

    return (
        <div>
            <p>Count: {count}</p>
            <button onClick={() => setCount(count + 1)}>
                Increment
            </button>
            <button onClick={() => setCount(prev => prev - 1)}>
                Decrement (functional update)
            </button>
        </div>
    );
}
```

State Updates are Asynchronous

React batches state updates for performance.

```
// BAD: May not work as expected
function handleClick() {
    setCount(count + 1);
    setCount(count + 1);
    // Result: count + 1, not count + 2
}

// GOOD: Use functional updates
function handleClick() {
    setCount(prev => prev + 1);
    setCount(prev => prev + 1);
    // Result: count + 2
}
```

Rule: When new state depends on previous state, use the functional form.

Complex State with Objects

```
interface Filters {
    search: string;
    type: string;
    maxPrice: number;
}

function PropertyFilters() {
    const [filters, setFilters] = useState<Filters>({
        search: '',
        type: 'all',
        maxPrice: 1000000
    });

    const updateFilter = (key: keyof Filters, value: string | number) => {
        setFilters(prev => ({
            ...prev,        // Spread previous state
            [key]: value    // Update specific field
        }));
    };

    return <input onChange={(e) => updateFilter('search', e.target.value)} />;
}
```

useEffect: Side Effects

Effects let you run code after render.

```
import { useEffect, useState } from 'react';

function PropertyList() {
    const [properties, setProperties] = useState<Property[]>([]);

    useEffect(() => {
        // Runs after every render (no dependency array)
        // Runs once on mount (empty array [])
        // Runs when dependencies change ([dep1, dep2])

        async function fetchProperties() {
            const data = await api.getProperties();
            setProperties(data);
        }

        fetchProperties();
    }, []); // Empty array = run once on mount

    return <div>{/* render properties */}</div>;
```

Effect Cleanup

Some effects need cleanup (subscriptions, timers).

```
useEffect(() => {
    // Setup
    const timer = setInterval(() => {
        console.log('Tick');
    }, 1000);

    // Cleanup function (runs before next effect or unmount)
    return () => {
        clearInterval(timer);
    };
}, []);

// With event listeners
useEffect(() => {
    const handleResize = () => setWidth(window.innerWidth);
    window.addEventListener('resize', handleResize);

    return () => window.removeEventListener('resize', handleResize);
```

useMemo: Expensive Calculations

Memoize computed values to avoid recalculation.

```
import { useMemo } from 'react';

function PropertyList({ properties, filters }: Props) {
    // Only recalculates when properties or filters change
    const filteredProperties = useMemo(() => {
        return properties
            .filter(p => p.type === filters.type || filters.type === 'all')
            .filter(p => p.price <= filters.maxPrice)
            .filter(p => p.title.toLowerCase().includes(filters.search));
    }, [properties, filters]);

    return (
        <div>
            {filteredProperties.map(p => <PropertyCard key={p.id} property={p} />)}
        </div>
    );
}
```

useCallback: Stable Function References

Memoize functions to prevent unnecessary re-renders.

```
import { useCallback } from 'react';

function PropertyList() {
    const [properties, setProperties] = useState<Property[]>([]);

    // Without useCallback, this creates a new function every render
    // causing child components to re-render unnecessarily
    const handleDelete = useCallback((id: string) => {
        setProperties(prev => prev.filter(p => p.id !== id));
    }, []); // No dependencies = stable reference

    return (
        <div>
            {properties.map(p => (
                <PropertyCard key={p.id} property={p} onDelete={handleDelete} />
            ))}
        </div>
    );
}
```

4. Component Patterns

Shadcn UI: Component Library

Pre-built, customizable components based on Radix UI.

```
# Installation
npx shadcn@latest init
npx shadcn@latest add button card input
```

```
// Usage
import { Button } from '@/components/ui/button';
import { Card, CardHeader, CardTitle, CardContent } from '@/components/ui/card';

function PropertyCard() {
    return (
        <Card>
            <CardHeader>
                <CardTitle>Beach House</CardTitle>
            </CardHeader>
            <CardContent>
                <Button variant="outline">View Details</Button>
            </CardContent>
        </Card>
```

Form Handling with React Hook Form

Uncontrolled forms with validation using React Hook Form + Zod.

```
import { useForm } from 'react-hook-form';
import { createPropertySchema, CreatePropertyInput } from '@/types/property';

function PropertyForm({ onSubmit }: Props) {
    const { register, handleSubmit, formState: { errors } } = useForm<CreatePropertyInput>({
        resolver: async (values) => {
            const result = createPropertySchema.safeParse(values);
            if (result.success) return { values: result.data, errors: {} };

            const errors = result.error.issues.reduce((acc, issue) => ({
                ...acc,
                [issue.path[0]]: { type: issue.code, message: issue.message }
            }), {});
            return { values: {}, errors };
        },
        defaultValues: { title: '', price: 0, propertyType: 'apartamento' }
    });

    return (
        <form onSubmit={handleSubmit(onSubmit)}>
            <input {...register('title')} />
            {errors.title && <p>{errors.title.message}</p>}
            <button type="submit">Save</button>
        </form>
```

Form Validation

Validate inputs before submission.

```
function PropertyForm() {
    const [errors, setErrors] = useState<Record<string, string>>({});

    const validate = (): boolean => {
        const newErrors: Record<string, string> = {};

        if (!formData.title.trim()) {
            newErrors.title = 'Title is required';
        }
        if (Number(formData.price) <= 0) {
            newErrors.price = 'Price must be positive';
        }

        setErrors(newErrors);
        return Object.keys(newErrors).length === 0;
    };

    const handleSubmit = (e: React.FormEvent) => {
        e.preventDefault();
        if (validate()) {
            // Submit form
        }
```

React Router: Navigation

Client-side routing for SPAs.

```
import { BrowserRouter, Routes, Route, Link } from 'react-router-dom';

function App() {
    return (
        <BrowserRouter>
            <nav>
                <Link to="/">Home</Link>
                <Link to="/new">Add Property</Link>
            </nav>

            <Routes>
                <Route path="/" element={<HomePage />} />
                <Route path="/new" element={<NewPropertyPage />} />
                <Route path="/property/:id" element={<PropertyDetailPage />} />
                <Route path="*" element={<NotFound />} />
            </Routes>
        </BrowserRouter>
    );
}
```

Dynamic Routes & Params

Access URL parameters in components.

```
import { useParams, useNavigate } from 'react-router-dom';
import { getPropertyById } from '@/lib/storage';

function PropertyDetailPage() {
    const { id } = useParams<{ id: string }>();
    const navigate = useNavigate();
    const [property, setProperty] = useState<Property | null>(null);

    useEffect(() => {
        if (id) {
            const found = getPropertyById(id);
            setProperty(found);
        }
    }, [id]);

    if (!property) return <p>Property not found</p>;

    return (
        <div>
            <h1>{property.title}</h1>
            <button onClick={() => navigate('/')}>Go Home</button>
        </div>
```

Local Storage Persistence

Save state to localStorage for persistence.

```
// lib/storage.ts
export function saveToStorage<T>(key: string, data: T): void {
    localStorage.setItem(key, JSON.stringify(data));
}

export function loadFromStorage<T>(key: string, defaultValue: T): T {
    const stored = localStorage.getItem(key);
    return stored ? JSON.parse(stored) : defaultValue;
}

// Usage in component
const [properties, setProperties] = useState<Property[]>(() =>
    loadFromStorage('properties', sampleProperties)
);

useEffect(() => {
    saveToStorage('properties', properties);
}, [properties]);
```

5. Deep Dive

1. Virtual DOM

React uses a virtual representation of the DOM for efficiency.

```
┌──────────────────────────────────────────────────────────────────────────┐
│                                Virtual DOM                               │
├──────────────────────────────────────────────────────────────────────────┤
│                                                                          │
│   [State Change]                                                         │
│        │                                                                 │
│        ▼                                                                 │
│   [New Virtual DOM] ◀─── React creates new tree                          │
│        │                                                                 │
│        ▼                                                                 │
│   [Diff Algorithm] ◀─── Compares old vs new                              │
│        │                                                                 │
│        ▼                                                                 │
│   [Minimal Updates] ◀─── Only changed nodes updated                      │
│        │                                                                 │
│        ▼                                                                 │
│   [Real DOM]                                                             │
│                                                                          │
```

2. Reconciliation & Keys

How React decides what to update.

Without Keys:

```
// React can't tell which item changed
<ul>
    <li>Item A</li>
    <li>Item B</li>  {/* If removed, React might update wrong item */}
    <li>Item C</li>
</ul>
```

With Keys:

```
// React tracks each item by key
<ul>
    {items.map(item => (
        <li key={item.id}>{item.name}</li>  // Stable identity
    ))}
</ul>
```

3. React Strict Mode

Development tool for finding potential problems.

```
// main.tsx
import { StrictMode } from 'react';

createRoot(document.getElementById('root')!).render(
    <StrictMode>
        <App />
    </StrictMode>
);
```

What it does:

- Renders components twice (in dev) to catch side effects
- Warns about deprecated APIs
- Detects unexpected side effects in render phase
- **Note:** This is why effects run twice in development!

4. Component Re-rendering

Understanding when components re-render.

Triggers:

1. State changes (`setState`)
2. Props change (parent re-renders)
3. Context value changes

Optimization:

```
// React.memo prevents re-render if props unchanged
const PropertyCard = React.memo(function PropertyCard({ property }: Props) {
    return <div>{property.title}</div>;
});

// Only re-renders if property object reference changes
```

5. Vite + React

How Vite serves React applications.

```
┌──────────────────────────────────────────────────────────────────────────┐
│                                Vite Dev Server                           │
├──────────────────────────────────────────────────────────────────────────┤
│                                                                          │
│   Browser                                                                │
│       │                                                                  │
│       │  Request: /src/App.tsx                                           │
│       ▼                                                                  │
│   Vite                                                                   │
│       │                                                                  │
│       ├── Transform JSX → JavaScript (esbuild)                           │
│       ├── Transform TypeScript → JavaScript                              │
│       ├── Inject HMR client                                              │
│       │                                                                  │
│       ▼                                                                  │
│   Browser receives transformed module                                    │
│                                                                          │
│   [File Change] ──▶ HMR Update ──▶ Page updates without refresh          │
│                                                                          │
```

6. Challenge Lab

Practice & Application

Part 1: Property Comparison

Context:

Users want to compare multiple properties side-by-side before making a decision.

Your Task:

Implement a comparison feature that:

- Allows selecting up to 3 properties for comparison
- Shows a comparison table with key metrics
- Highlights the best value in each category
- Provides a way to remove properties from comparison

Files to Create/Modify:

- `src/components/CompareButton.tsx`
- `src/pages/ComparePage.tsx`

Part 1: Definition of Done

Criteria	Description
Selection UI	Checkbox or button to select properties
Max 3 limit	Cannot select more than 3 properties
Comparison table	Side-by-side view with property details
Metrics shown	Price, bedrooms, bathrooms, area, price/sqm
Best highlighted	Lowest price, highest area highlighted
Remove from compare	Can deselect from comparison view
Empty state	Message when no properties selected

Part 2: Image Gallery

Context:

Properties should support multiple images with a gallery view on the detail page.

Your Task:

Implement an image gallery that:

- Shows thumbnail grid of all property images
- Opens full-screen modal on click
- Supports keyboard navigation (arrow keys, escape)
- Shows image counter (e.g., "3 of 10")

Files to Create/Modify:

- `src/components/ImageGallery.tsx`
- `src/components/ImageModal.tsx`

Part 2: Definition of Done

Criteria	Description
Thumbnails grid	Property images shown as clickable thumbnails
Modal opens	Clicking thumbnail opens full-screen view
Navigation arrows	Left/right buttons to navigate images
Keyboard support	Arrow keys and Escape work
Image counter	Shows current position (e.g., "3 of 10")
Close button	X button to close modal
Backdrop click	Clicking outside image closes modal

Resources & Wrap-up

Resources

React

- React Documentation
- React Tutorial (Tic-Tac-Toe)
- Thinking in React
- React Hooks Reference

TypeScript + React

- React TypeScript Cheatsheet
- Total TypeScript: React

Shadcn UI

- Shadcn UI Documentation
- Shadcn UI Components

Recommended Articles

React Fundamentals

- A Complete Guide to useEffect - Dan Abramov
- Before You memo() - Dan Abramov
- React re-renders guide - Developer Way

Performance

- Optimizing Performance - React Docs
- useMemo and useCallback - Kent C. Dodds
- When to useMemo and useCallback - Kent C. Dodds